Jiawen (Jarvan) Wang

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PROFILE

Master of Information Technology student at the University of Technology Sydney seeking AI, data analytics, machine learning, or related technology internships. Project experience across sports analytics, computer vision, NLP, and healthcare prediction, using Python, SQL, Excel and Tableau, visualization, cross-validation, and metric-based evaluation.

CORE STRENGTHS

- Builds end-to-end workflows for data preparation, analysis, feature engineering, modeling, visualization, and error review.
- Evaluates models with Accuracy, F1-score, ROC-AUC, confusion matrices, validation curves, and cross-validation.
- Uses Python, SQL, Excel, Tableau, Scikit-learn, PyTorch, and Jupyter/Colab notebooks for reproducible analysis and experiments.

EDUCATION

University of Technology Sydney 2025-Present
Master of Information Technology | Coursework: Database, Data Analytics, Advanced Data Analytics Algorithms, Project Management, Software Development

Huazhong Agricultural University 2019-2023
Bachelor of Biotechnology | Coursework: Microbiology, Cell Biology, Genetics, Biochemistry, Genetic Engineering, Pharmacology

PROJECTS

Deep Learning for Image Classification and Object Detection
PyTorch, ResNet18, Faster R-CNN, SSD300

- Built PyTorch pipelines for image classification and object detection with **ResNet18, Faster R-CNN, and SSD300**, using annotations and predictions for model debugging.
- Compared baseline and customized ResNet18 architectures using training/validation loss and accuracy curves to assess convergence and overfitting.

Handwritten Mathematical Symbol Recognition
Python, Scikit-learn, OpenCV, HOG, LBP

- Engineered HOG, LBP, and raw-pixel feature representations for a multi-class symbol recognition workflow.
- Trained SVM and ANN classifiers; compared confusion matrices and accuracy metrics to identify misclassification patterns.

Cardiovascular Disease Prediction
Python, Scikit-learn, Pandas, NumPy

- Cleaned and engineered features from **56,000+ medical records**, including missing-value handling and structured preprocessing.
- Built Logistic Regression, Random Forest, and KNN models; compared Accuracy, F1-score, and ROC-AUC to select the strongest model.

T20 Win Probability & Player Contribution
Python, Pandas, Scikit-learn, Random Forest

- Built a reproducible ball-by-ball pipeline across **4,975 T20I matches** and **1.16M delivery events** with chronological train/validation/test splits.
- Engineered leakage-safe game, player, team, recent-form, and venue features; selected Random Forest by log loss and converted win-probability changes into player NWC.

TECHNICAL SKILLS

Analytics & BI: Excel, Tableau, SQL, Pandas, NumPy, Matplotlib, Seaborn, data cleaning, visualization, reporting

Machine Learning & AI: Scikit-learn, PyTorch, Random Forest, CNN, ResNet18, model evaluation, feature engineering, cross-validation

Programming, Cloud & Tools: Python, Linux/Unix command line, Git/GitHub, Jupyter Notebook, Google Colab, VS Code, OpenCV, TF-IDF, basic AWS exposure | English, Mandarin Chinese